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Towards roll-to-roll printed batteries based on organic electrodes for printed electronics applications

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ABSTRACT

There is a growing interest in smart packaging applications and small IoT devices realized as printed electronic devices. Because of their electrical nature, these printed devices require small rechargeable energy sources for their repeatable functionality. For these reasons, the printing of rechargeable batteries alongside the main printed electronics features is welcome, especially if it is realized by the roll-to-roll (R2R) process. A reliable R2R printing process is an advantage and it is a solution overcoming technology difficulties related with the conventional approach of bonding pouch cells to these devices. R2R printing process allows battery manufacturing in the same printing line, at the same substrate, with proper shape and required capacity. Because most of the printed electronic devices are manufactured by R2R printing processes we present a study that is focused on the development of electrode materials towards fully R2R printed secondary batteries based on easily accessible materials and printed by robust and scalable technology. We demonstrate a fully R2R printed secondary battery with open-circuit voltage (OCV) higher than 2.0 V where an organic-based anode and a cathode without Co and Ni are used. The R2R printed secondary battery could be assembled at ambient conditions, which is important for the development of a cheap industrial R2R manufacturing process without inert atmosphere requirements. The R2R printed battery assembled at ambient conditions provides 100 cycles with a capacity of over 50 mAh/g. In the presented study, it was demonstrated that printed PAQS electrodes could be used also in the Na-ion batteries with a capacity of over 120 mAh/g.

1. Introduction

Currently, there is an increasing interest in flexible electronics applications, especially in the area of smart packaging. A variety of intelligent devices like displays [1], radio-frequency identification (RFID) and smart labels [2], medical sensors [3], manufactured by printing and coating technologies requires a source of electric energy. Although these printed electronic devices have a relatively short life cycle, there are requests for repeated usage. For this reason, rechargeable batteries in these devices are welcome. Because printed electronics structures are mainly manufactured by printing techniques, it is an advantage to manufacture batteries in the same printing line/process together with the main device. Nowadays, many processes are tested for the creation of printed battery systems such as flexographic, screen-printing, imprint lithography, spray deposition, inkjet printing, or selective laser sintering [4-7].

In our study, we are presenting this research study focused on the development of a fully printed battery with OCV higher than 2 V, which can serve as an energy supply for many types of printed/hybrid electronic devices. At the same time, the research is focused on the use of easily available and environmentally friendly electrode materials. We use an organic poly(1,5-anthraquinonyl sulfide) (PAQS) material at the side of the anode, which enables easier recycling and we use a $LiMn_2O_4$ (LMO) material for cathode, which does not contain Co and Ni. The presented study covers the development of electrode materials up to fully R2R printed secondary batteries with OCV higher than 2.0 V manufactured at ambient conditions with a relative humidity of 20% RH and their testing.

There are a lot of studies focused on quasi solid-state batteries based on organic materials [8-12] in the literature during recent years.

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Lithium, sodium, or other ions are used in electrolytes to ensure the proper electrochemistry of these cells. The advantage of organic compounds is that selected characteristics (solubility, molecular weight, number of electron transitions, etc.) can be modified by a suitable synthetizing procedure. Various types of secondary batteries using electrodes based on organic compounds are described in the literature. Organic electrode materials commonly exhibit multi-electron transitions during the charging/discharging cycles, enabling high specific capacity typically in the range of 300–800 mAh/g [13]. Although many organic cathodes/electrodes exhibit high specific capacity in solid-state batteries, negative properties such as cycling instability, low electrical conductivity, or high solubility in the electrolyte are still present. One of the most important properties is low solubility which is crucial for stable battery behaviour and high specific capacity [8-12]. The low solubility problem can be solved by various approaches, such as the appropriate structure substitution, polymeric material formation from monomers, using electrolytes based on orthogonal solvents, or using a filler that adsorbs the dissolved materials on its surface. One of the mentioned approaches is the use of quasi-solid electrolytes [13,14] but these batteries often suffer from low ion mobility in electrolytes [15]. The electrochemistry of organic compound electrodes is often based on the reduction of keto groups present to the hydroxy group. This electrochemistry phenomenon is present for the simplest quinone derivatives, which are usually represented by simple quinoid structures such as

1,4-benzoquinone, 9,10-phenanthroquinone [16,17] or 9,10-anthraquinone [18]. The last one was used as a solid cathode electrode in combination with mesoporous carbon [12] in a cell with a specific capacity of 222 mA.h/g. The next group of materials are substituted quinones such as

2,5-dimethoxy-1,4-benzoquinone [10] with the specific capacity of 312 mA.h/g (98% of theoretical), 2,3,5,6-tetrachloro-1,4-benzoquinone, 2,3-dichloro-5,6-dicyano-1,4-benzoquinone (DDQ) [14],

2,5-dihydroxybenzoquinone [14], 2,3,5,6- 2,4-trihydroxyantraquinone [19]. Higher cycling stability was exhibited by sulfo-substituted materials [20] or polymeric materials prepared by the Phillips method from 9,10-anthraquinone monomer by the reaction with sodium sulphide [9,10,21]. Poly(anthraquinonyl sulfide) [9] shows the specific capacity of 198 mA.h/g (88% of theoretical) during the first cycle and 178 mAh/g after 200 cycles. This polymeric material also shows only 18% drop of capacity between the loads of 0.25C and 2.5C. Interesting results were published by Li et al. [22] for poly(1,4-anthraquinone). He manufactured a PAQS/air battery with 253 mAh/g at 1C with an OCV of 0.7 V. Bitenc et al. used PAQS [23] for magnesium battery where he obtained a specific capacity of 150 mAh/g. High cycling stability also provides polyquinonimines synthesized by the reaction of 2,6-diaminoanthraquinone with dianhydrides [24]. The synthesized polymer exhibited a specific capacity of 210 mA.h/g and it showed a specific capacity of 140 mA.h/g after 2000 cycles during cycling at 5C.

The inks for electrode manufacturing are mostly composites, which usually contain a film-forming polymer (polyvinylidene difluoride (PVDF), polytetrafluoroethylene (PTFE), etc.), conductive carbon in various forms (active carbon [25], graphite [26], carbon nanotubes [27]), which ensures charge transportation and partially prevents dissolution of the electrode materials, the organic electrode material itself, and the solvent allowing to set the proper rheology of ink. We can find some electrode materials in literature where the electrically conductive parts are based on conductive polymers [28-31] which provide electrochemical activity themselves and thus contribute to an increased specific capacity.

Studies focused on printed secondary batteries can be found in the literature, but most of them are focused on the sheet-to-sheet printing process. Lawes et al. [32] prepared an anode based on silicon nanoparticles, carbon black, and binder system based on various polymers (poly(3,4-ethylenedioxythiophene), polystyrene sulfonate (PEDOT:PSS), polyvinylpyrrolidone (PVP), carboxymethyl cellulose (CMC), sodium alginate) by the ink-jet printing technique. The same printing technique used Zhao et al. for the fabrication of electrodes for lithium-ion batteries (LIB) based on SnO_2 [33], $Li_4Ti_5O_{12}$ (LTO) [34], or $LiCoO_2$ (LCO) [35]. The screen printing technique was used for the manufacturing of cathodes based on $LiCoO_2$ [36], $LiFePO_4$ (LFP) [31], or cathode for Li-O_2 batteries [37].

To the best of our knowledge, no particular study investigating roll-to-roll printed secondary batteries with organic-based electrodes with voltage over 2 V has been published. Hence, the purpose of this study was to demonstrate the suitability of an organic electrode based on PAQS in combination with an LMO cathode for R2R manufacturing of a secondary battery at ambient conditions. The performance of the PAQS electrodes manufactured in an inert atmosphere was also evaluated depending on used electrolytes, conductive fillers and towards their usage in LIB and sodium-ion batteries (SIB).

2. Experimental

2.1. Materials

Synthetized poly(1,5-anthraquinonyl sulfide) (PAQS) was used as electrode material in this study. The PAQS was synthesized according to Song [9].

2.2. Synthesis of poly(1,5-anthraquinonyl sulphide) - PAQS

250 ml of N-methylpyrrolidone was poured into a four-mouth flask, with a thermometer, stirrer, and a reflux condenser with a distillation head, placed on a heating nest. 0,15 mol of 1,5-dichlorantrachinon (Merck, purity 96%) and 0,15 mol of sodium sulphide hydrate (Fluka, content 59%) were added. Flask content was stirred for 5 h at 200 °C. Distillation took place at 165 °C. 50 ml of the distillate containing water was collected in the distillation head. The heating was then switched off. 500 ml of water was poured into the flask after cooling down to 100 °C. It was stirred overnight until the next day. The fine brown suspension was filtered through a frit S4. 300 ml of water was cleaned using a filter. The filter cake was dried for 20 hours at 100 °C. The yield was 34 g (95%) of poly-anthraquinone sulfide.

2.3. Ink formulation for stencil and screen-printing technique

The film-forming polymer for waterborne printing ink formulation was CMC (Sigma Aldrich) with Mw 250 000 and degree of substitution 1.2. Demi water was used as the solvent. Ink conductive components were Super C45 (Timcal Ltd.) with particle size 100–200 nm and specific surface area 45 m^2/g and graphene nanoplatelets (GNP) made from expanded graphite with average particle size 5 and specific surface area 150 m^2/g. Waterborne printing ink was as follows: 7% wt. of CMC solution in demi water then Super C45 or GNP was added to the polymer solution (6% wt.) and homogenized carefully by an overhead stirrer (Heidolph Hei-TORQUE Precision 100). In the next step, the milled PAQS was added to the ink (24.5% wt.). Additional styrene-butadiene rubber (SBR (Gelon)) dispersion in the amount of 4% wt., humectants, and surface tension additives (glycols) in the amounts of 10% wt were added. The ink was homogenized for 12 h using the overhead stirrer. The viscosity of the ink was 34.4 Pa.s for the Super C45 based ink and 42.6 Pa.s for the GNP-based ink (Brookfield DV-*I*+, 100 RPM).

2.4. Printed electrodes and batteries prepared by stencil/screen-printing technology

The electrodes were printed on an Al foil with up to 11.4 mg of active material per cm^2 for the study of electrode layers in an Ar atmosphere. A Stencil with a thickness of 175 μm was used for printing on the Cu and Al foils. Stencil printing was done using a semiautomatic screen-printing machine Ekra E1, with a printing speed of 70 mm/s. The printed layers were dried at 70 °C for 60 min (Memmert hot air oven UF75).

Discs with a diameter of 18 mm were cut out from the printed layer on an Al foil and subsequently calendered by the pressure of 300 kg.cm^{-2}. Disc electrodes were subsequently dried at 100 °C in a vacuum. The dried electrodes were transferred into a Jacomex glove box with an argon atmosphere with less than 1 ppm of oxygen and water content. Additional drying was performed in the glove box oven at 60 °C. In the next step, the discs were inserted into an electrochemical cell El-Cell© ECC-Std. A metal lithium disc was used as the counter electrode for lithium-based cells and electrolytes were 1 mol.l^{-1} $LiPF_6$ (lithium hexafluorophosphate) in EC (ethylene carbonate):DMC (dimethyl carbonate) in the ratio of 1:1 w/w or 0.25 mol.l^{-1} $LiNO_3$ (lithium nitrate) + 0.7 mol l^{-1} LiTFSI (lithium bis(tri-fluoromethanesulfonyl)imide) in DME (1,2-dimethoxyethane) and DOL (1,3-dioxolane) in the ratio of 2:1 v/v. A metal sodium disc was used as the anode for sodium-based cells. The electrolytes in this case were a mixture of 1 mol.l^{-1} $NaClO_4$ (Sodium perchlorate) in EC:DMC in the ratio 1:1 w/w or 0.8 mol.l^{-1} $NaPF_6$ in DME and DOL in the ratio 2:1 v/v. The electrolyte was soaked into a glass fibre separator in all cases.

2.5. R2R screen-printed secondary batteries

Fully printed sandwich-type batteries were printed in the R2R printing regime. The active area of cells was 3.5 × 3.5 cm. Designed 12 cm^2 active area allowed to prepare battery with capacity 4 mAh which is suitable for powering of small devices as NFC smart labels, electrochromic display, etc. Cells consisting of PAQS as anode and LMO (Sigma Aldrich) as cathode were printed by the R2R process. The GM SC330 R2R printing machine was used for printing. Polyethylene terephthalate (PET) of 150 mm width (100 μm PET DuPont Teijin Films Melinex ST506) was used as a printing substrate. In the first step, the silver-based current collectors (DuPont 5029) were printed (SAATI SAATILENE HIBONDplus mesh with 77 thr./cm coated by a KIWO emulsion) at a printing speed of 1.2 m/minute and dried by hot air (120 °C). The resulting layer thickness was 10.3 μm. The second layer was based on a solvent-borne ink formulation developed by us based on carbon composite consisting of graphite, carbon black, and a nonpolar polymer binder with solid content 44.5 wt.%. The viscosity of the ink was 14.2 Pa.s (Brookfield DV-*I*+, 100 RPM). This layer ensured the protection of the silver current collector against the electrolyte. The printing speed was 1.2 m/minute and it was dried by hot air (120 °C). The printing form was based on the PES SAATI SAATILENE HIBOND plus mesh with 77 thr./cm. The thickness of dried carbon layer was 11.8 μm. The third layer was the anode layer consisting of the PAQS ink described above. This layer was printed in two passes using a printing form with the SAATI SAATILENE HIBOND plus 24 thr./cm at a printing speed of 0.76 m/minute and with drying at 100 °C. The thickness of dried PAQS layer was 51.3 μm. The fourth layer – a cathode based on LMO ink formulation (viscosity 9.3 Pa.s (Brookfield DV-*I*+, 100 RPM)) was printed with a mesh SAATI SAATILENE HIBOND plus 43 thr./cm in the same way as the anode but only in one pass. The thickness of dried LMO layer was 16.4 μm. About 40 m of batteries were printed, which means more than 150 samples of batteries. Selected samples were assembled into cells, the cells were folded and glued using double-sided adhesive tape and micro-dispersed epoxy glue resistant to the electrolyte. Gel electrolyte based on 1 M LiTFSI in ionic liquid (IL) was stencil-printed on cathode and anode before cell gluing. The glass fibre-based separator (4.2 × 4.2 cm) was used to prevent electrodes´ shortcuts. The process was developed for production at constantly set ambient conditions; therefore, cell assembly was performed in laboratory conditions with 20% of RH and 20 °C. The assembled cells were then gently calendered to unify the gap and amount of electrolyte between the electrodes.

2.6. Determination of the output characteristics

The electrochemical cells assembled in an argon atmosphere were characterised using a potentiostat VMP3 (Bio-Logic). A scanning electron microscope (SEM) TESCAN VEGA3 XMU and a Bruker EDAX analyser were used to study the electrode surface before and after cycling. The cells which were printed in the R2R regime were characterised using a potentiostat ORIGALYS ORIGAFLEX – OGF500 with an ORIGAFLEX-OGFPWR power source. The cells were also characterised using cyclers NEWARE 5V50MA. The viscosity of ink formulation was estimated by viscometer Brookfield DV-*I*+. The layer thickness of various layers was measured by using mechanical profilometer KLA Tencor P-7.

3. Results and discussion

3.1. The influence of the conductive filler and electrolyte solvents on PAQS electrode performance in LIB cells

The ink formulation was optimised to obtain good printing behaviour, good electrochemical properties of the printed layers, and proper mechanical properties of electrode layers towards flexible electronics applications. The printed electrodes were based on synthesized PAQS. The first set of experiments was focused on testing of PAQS electrodes performance in a Li-ion-based system depending on the conductive filler with a water-based system consisted from CMC and SBR as a filmforming and binder system. Two types of conductive fillers were used, where the first type of layers (PAQS-C45) was based on Super C45 as a typical conductive carbon black material with small-sized globular shaped particles, which have high conductivity and small specific surface. The second layer (PAQS-GNP) was based on GNP fabricated from expanded graphite. These GNP particles are flakes that exhibit a high dimension aspect ratio shape. These particles have high conductivity and it was expected that they will enable good charge transfer within the electrode. Two types of solvent-based electrolytes were used for the estimation of the electrochemical characteristics of these layers. The first combination of carbonates-based solvents was EC:DMC and the second combination of glymes-based solvents was DME:DOL. The first step after the assembly of the electrochemical test cells was an electrochemical impedance spectroscopy (EIS) analysis, 5 cycles of cyclic voltammetry were then performed at various scan rates (5, 1, 0.5, and 0.1 mV/s) for the formatting of the cells and estimation of the basic parameters of the electrodes.

It is evident from Fig. 1, that the cells with the glyme-based electrolyte exhibited much higher activity than the electrodes with the carbonate-based electrolyte. This behaviour has similar trends for both types of conductive fillers. The anodic/cathodic peak for PAQS-C45 and PAQS-GNP has the same maximum of 2.50/2.08 V when the glyme-based electrolyte is used. It was approximately 2.62/2.17 V for the carbonates at the scan rate of 5 mV/s. The difference in maxims of the anodic/cathodic peak is 0.42 V for PAQS-C45 cell with the glyme-based electrolyte for CV at 5 mV/s. The difference of anodic/cathodic peak is quite similar ~ 0.45 V for the carbonate-based electrolyte cell at the same scan rate. However, for CV at a lower scan rate of 0.1 mV/s, the PAQS-C45 cell with glyme-based electrolyte exhibited a difference in maxims of anodic/cathodic peaks of 0.26 V and the difference was 0.33 V in the case of the PAQS-C45 cell with the carbonate-based electrolyte. It is also obvious, that anodic and cathodic peaks are shifted about 0.12 V to the higher potential for all cells with the carbonate-based electrolyte. Current density during CV was approximately 3 times higher for cells with the glyme-based electrolyte and both types of conductive fillers. These results were caused by the higher internal resistance of the cells with the carbonate-based electrolyte. This difference is probably given by the worse wetting or penetrating capability of the carbonate-based electrolyte towards the electrode structure. It is evident from data sets of all scan rates that with the increasing scan rate value, the difference between anodic and cathodic peaks is increasing and anodic and cathodic peaks are shifted towards the lower/higher potential for all cells. This phenomenon is typical and it is given by the capability of cells to receive charge at higher current loads. It is clear that, in terms of the

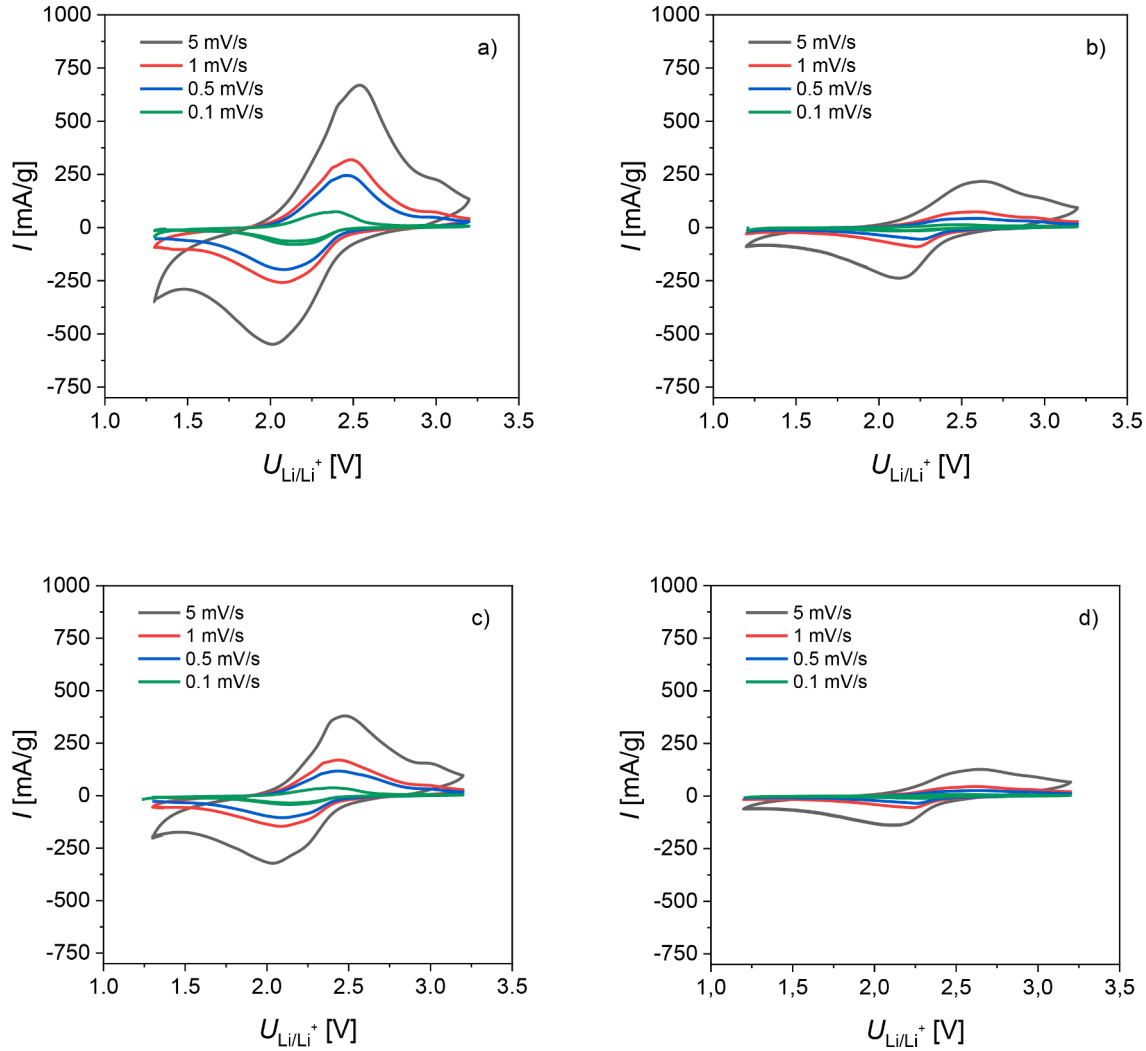

Fig. 1. (A) Cyclic voltammetry of PAQS-C45 layers at various scan rates (0.1–5 mV/s) for glyme-based Li electrolyte (B) Cyclic voltammetry of PAQS-C45 layers at various scan rates (0.1–5 mV/s) for carbonate-based Li electrolyte (C) Cyclic voltammetry of PAQS-GNP layers at various scan rates (0.1–5 mV/s) for glyme-based Li electrolyte (-D) Cyclic voltammetry of PAQS-GNP layers at various scan rat.

influence of the conductive fillers on the performance of the cells, the cell with PAQS-C45 is more active because the maximal current density for the scan rate 5 mV/s was approximately 680 mA/g, where it was 380 mA/g for the PAQS-GNP electrode with the glyme-based electrolyte. The result was similar in the case of the carbonate-based electrolyte: 240 mA/g in the case of the PAQS-C45 and 140 mA/g for the PAQS-GNP.

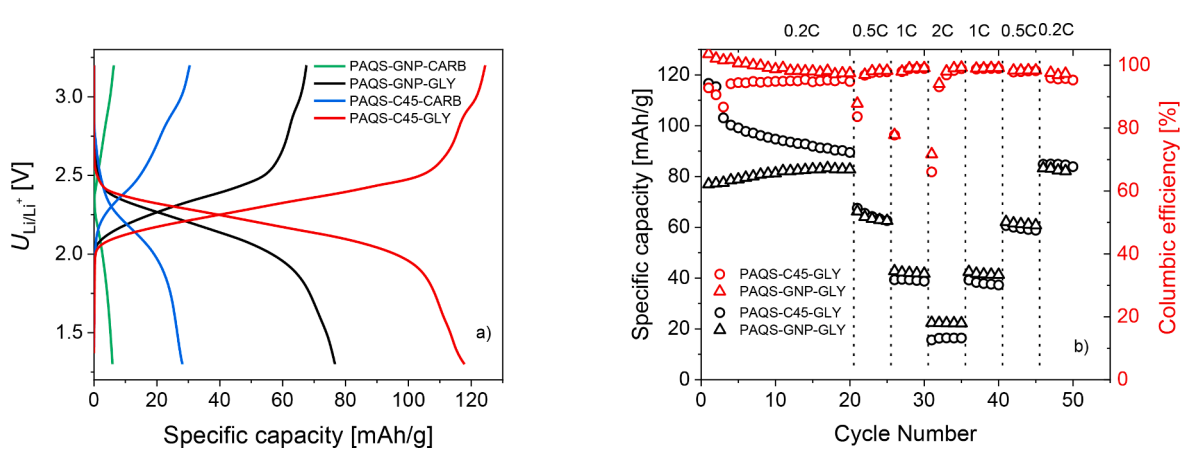

Fig. 2. (A) Galvanostatic cycling at a cycling rate of 0.2C for glymes- and carbonates-based type Li electrolytes for electrodes with conductive fillers based on Super C45 and GNP, (B) Galvanostatic cycling at various cycling rates for glymes- and carbonates-based type Li electrolytes for electrodes with conductive fillers based on Super C45 and GNP.

This could be caused probably by lower barrier properties of globular Super C45 particles in comparison to the flake nature of GNP particles and so easier diffusion of ions into layers with Super C45.

Galvanostatic cycling was performed in the next step. Here we can also observe a big difference in the performance of the electrodes assembled with carbonates and glymes-based electrolytes. It is evident from Fig. 2a that the cell with the carbonates-based electrolyte has much lower performance (5.9 mAh/g and 28.1 mAh/g) than with the glymes-based solvents (76.6 mAh/g and 124.4 mAh/g).

The cycling performance of the electrodes with glymes-based electrolyte at various C-rates is in Fig. 2b. It is obvious from the charts that the electrode with the Super C45 conductive filler enabled the highest specific capacity of about 120 mAh/g but the electrode with GNP has better results in terms of stability at higher C-rates (over 1C). The capacity of the electrode with GNP at a C-rate of 2C was higher (22.2 mAh/g) in comparison to the Super C45 based electrode (16.4 mAh/g). It seems that the GNP electrode has nearly the same capacity after 50 cycles (82.1 mAh/g) as the maximum after cell formatting (83.3 mAh/g). The higher capacity decrease of Super C45 could be caused by higher mechanical stress of the layers because of the globular shape and higher surface area of Super C45 particles compared with the flake shape of GNP. The high aspect ratio of the GNP particles shape caused a positive effect to charge transfer at higher current loads. In respect to the final application, the electrode with the Super C45 conductive filler was used because of its higher specific capacity.

In addition, an EIS analysis was performed before cycling and after 20 cycles at 0.2C. It is clear from Fig. 3 that the internal resistance of the cells with the carbonate-based electrolyte is higher. The figure also shows the equivalent circuit model used for all curves, while the values of *Re* (electrolyte resistance), Rs (resistance of the interfacial layer), and Rct (charge transfer resistance) are in Table 1 The charge transfer resistance of the PAQS-GNP-CARB electrode cell was 88 Ω and it was even higher in the case of the PAQS-C45-CARB electrode (290 Ω). On the other hand, the charge transfer resistances of the PAQS-GNP-GLY and the PAQS-C45-GLY electrodes were 9 Ω and 26 Ω, respectively. The difference is given by the higher conductivity of the glyme-based electrolyte which was 26.7 mS cm^{-1} in comparison to 14.2 mS cm^{-1} of the carbonate-based electrolyte. The next influence could be the different

Table 1
. Parameters of the equivalent circuit model for selected PAQS electrod.

Type of electrode	*Re* [Ω]		Rs [Ω]		Rct [Ω]	
	Before cycling	After cycling	Before cycling	After cycling	Before cycling	After cycling
PAQS-GNP-CARB	1.68		5.95		88.0	
PAQS-GNP-GLY	1.50	1.27	0.93	6.10	9.1	14.2
PAQS-C45-CARB	1.17	2.12	1.23	0.35	290.1	226.0
PAQS-C45-GLY	3.0	2.22	4.52	26.2	26.3	31.1

penetrating capabilities of electrolytes into the electrode structure. The charge transfer resistance of the PAQS-C45-CARB electrode after cycling decreased about 22% to 226Ω, which corresponds with the penetration/forming process during cycling. The charge transfer resistances of the PAQS-GNP-GLY and PAQS-C45-GLY electrodes increased about 55% to 14 Ω and about 19% to 31 Ω, respectively. Data obtained by EIS before cycling and after 20 cycles are in correlation with the results obtained during cycling at a higher C-rate where PAQS-GNP-GLY electrode showed higher capacity and stability, proving lower internal resistance of the GNP-based electrodes. The PAQS-GNP-CARB electrode in combination with the carbonate-based electrolyte did not even pass the cycling test. The PAQS-C45-CARB electrode exhibited the highest charge transfer resistance.

Because of the greater capacity and greater impedance stability, the electrode PAQS-C45 with Super C45 in combination with glyme-based electrolyte was chosen for a long-term cycling test at 0.2C. We can see from the obtained data that at the given C-rate, that the decrease of capacity after 100 cycles was about 16% from the initial specific capacity of 117 mAh/g. These properties are more than sufficient for application in flexible electronics where in many cases devices require stable functionality only in tens of cycles. It is obvious from the SEM

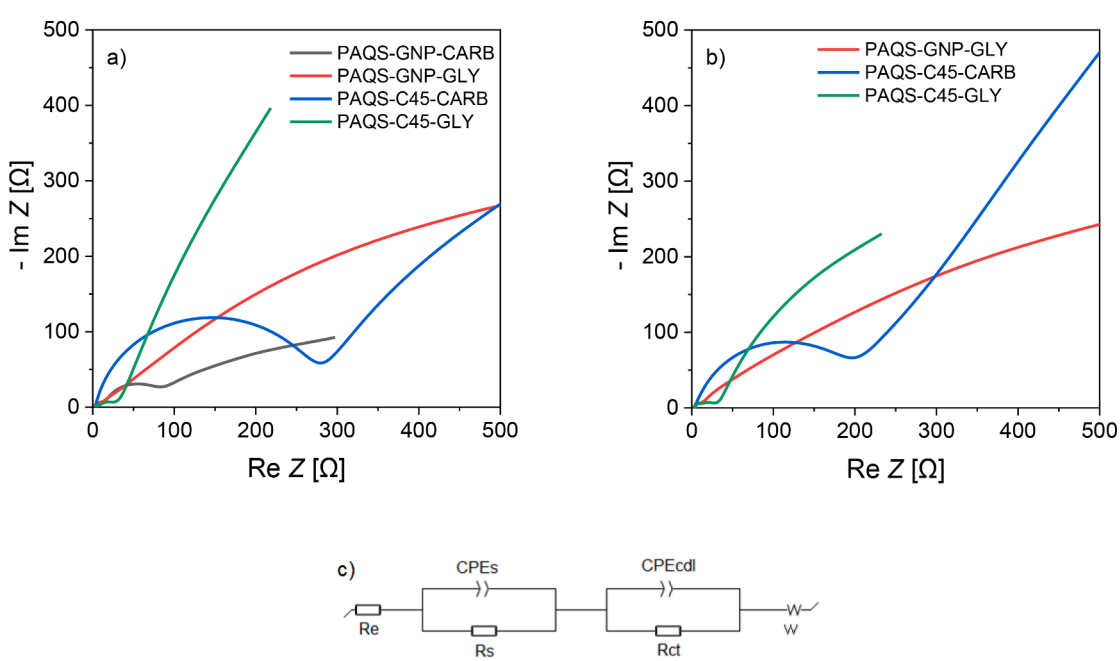

Fig. 3. (A) EIS analysis for cells with glymes- and carbonates-based type Li electrolytes and with electrodes with conductive fillers based on Super C45 and GNP – before cycling, (B) EIS analysis for cells with glymes- and carbonates-based type Li electrolytes and with electrodes with conductive fillers based on Super C45 and GNP – after 20 cycles at 0.2C, (C) equivalent circuit model.

image analysis (Fig. 4) that there are cracks in the cycled electrodes of PAQS which could affect the performance of the cells.

3.2. The printed Na-ion battery (SIB) using PAQS as a cathode

There is an increasing interest in SIB in various types of energy storage applications. Excluding lithium from energy sources (batteries) for smart packaging applications also seems to be interesting because of price, legislative restriction towards to LIB, etc. Alternatively, PAQS was used in magnesium batteries by Bitenc et al. [23]. For this reason, we tested the usability of PAQS electrodes for the SIB type of batteries. Electrochemical cells with metal sodium foil were assembled for PAQS based electrodes. It is obvious from the CV at the scan rate of 5 mV/s that the current density was nearly two times higher than for the equivalent cell (Fig. 5a). This can be caused by better Na^+ transport in electrolyte compared to Li^+. It is clear from the CV analysis of electrodes with both fillers (Super C45 and GNP) and electrolytes based on carbonates and glymes that the cells with carbonates-based electrolytes have a poor performance which was also concluded from the galvanostatic analysis.

We can see two cathodic peaks (1.77 and 2.16 V vs. Na/Na^+) and two anodic peaks (1.54 and 1.90 V vs. Na/Na^+) in the cyclic voltammetry results at a lower scan rate (in supplementary information). The oxidation peaks of electrodes with GNP are shifted about 0.06 V to higher values and reduction peaks are shifted about 0.04 V towards lower values. The presence of two anodic/cathodic peaks is obvious from the presence of two charge and discharge plateaus also in galvanostatic cycling results. Very similar specific capacities were obtained for the SIB cells for both types of fillers at 0.2C: 127.6 mAh/g for PAQS-C45 and 119.8 mAh/g for PAQS-GNP (Fig. 5b). This is a lower difference compared to the electrodes with glymes-based electrolyte with Li salt. The overall capacity for PAQS-C45 was similar to the results of the Li-based electrolyte equivalent. The OCV of SIB based on PAQS cathode exhibits overall lower energy density in comparison to LIB equivalent because Na/Na^+ has lower electrochemical redox potential. It is clear from the EIS analysis after 10 cycles that the cells exhibit lower internal resistance than their equivalents with Li glymes-based electrolyte (Fig. 6b). The change of impedance is given by the penetration of the electrolyte into the volume of the electrode during cycling.

3.3. Printed electrode for coin cell assembled at ambient conditions with electrolyte based on ionic liquid

Because the final goal was fully printed secondary batteries for smart packaging applications, the process requirement was battery manufacturing at constantly set ambient conditions (20 °C, 20% RH). The R2R printing process of the printed battery was developed for this purpose, including the development of specific layers geometry, their thicknesses, and some other specific engineering aspects. An important part of this research task was also the development of electrolyte, which enables the proper functioning of the battery even when the battery is assembled in air atmosphere with a humidity of 20% RH at 20 °C. The electrolyte based on 1 M LiTFSI dissolved in IL was found as a possibility for this purpose. The coin cell was used because of the ideal assembly enclosure. The PAQS electrode with Super C45 described before was used as an anode that was stencil-printed on a Cu foil. An LMO-based cathode was used. The ink formulation for stencil printing was developed again for a water-based binder system. The LMO was selected because it allowed assembling the battery at ambient conditions with high OCV and good overall stability of the whole cell. LMO is also a cathode material without the presence of Ni or Co which is another advantage for future smart packaging applications because it cannot be influenced by the overhyped market focused on electrode materials used in high energy density cells. The LMO ink was stencil-printed on an aluminium foil. The electrodes and glass fibre-based separator were dried (95 °C) in a hot air oven before the coin cell assembly and the cell was quickly assembled with the addition of 150 μL of the IL electrolyte. The coin cell was then closed in a crimping machine. A comparison of calendering (300 Kg/cm^2) influence was performed. The cells were cycled at various loads (Fig. 7a, b) and it is evident that initialization of the cells took place during the first ten cycles, including penetration of the electrolyte into the electrodes with the consequence of capacity increase. This behaviour is quicker for the non-calendered cell, probably due to the more open structure of the electrode. Interestingly, the cell with non-calendered electrodes has a lower capacity decrease at higher current loads which is caused by the more open structure of the electrode and easier ion transport. It is evident from the data that when the current load is lowered back to 0.1C, the capacity is recovered to the previous level of 88 mAh/g during the 70th cycle, which is higher than for the 25th cycle. This phenomenon is probably caused by the penetrating performance (wetting, low viscosity) of the IL electrolyte into electrode structure during cycling. This effect is also obvious in the comparison of 0.5C rates. When discharging characteristics are observed, the main plateau is apparent in the range of 2.1–1.5 V which is very useful for various types of devices in smart packaging applications.

A long-term stability test was performed for the coin cell consisting of the LMO cathode and the PAQS anode, filled by the IL-based electrolyte (Fig. 8). The cell was loaded by different current loads at the beginning, the current load was set to 0.5 C after the 50th cycle. The cell exhibits a capacity of 46.7 mAh/g after 500 cycles at this load (550 cycles in total), which is a 27% decrease of capacity in comparison with

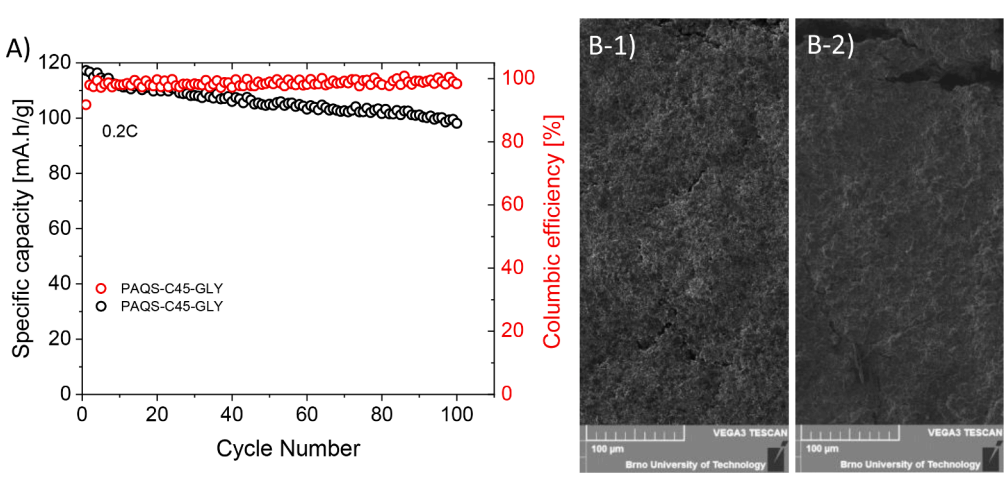

Fig. 4. (A) Long term galvanostatic cycling at cycling rate 0.2C for glymes-based type Li electrolyte for PAQS with conductive fillers based on Super C45, (B) SEM microscopy images before(B-1)/after cycling(B-2).

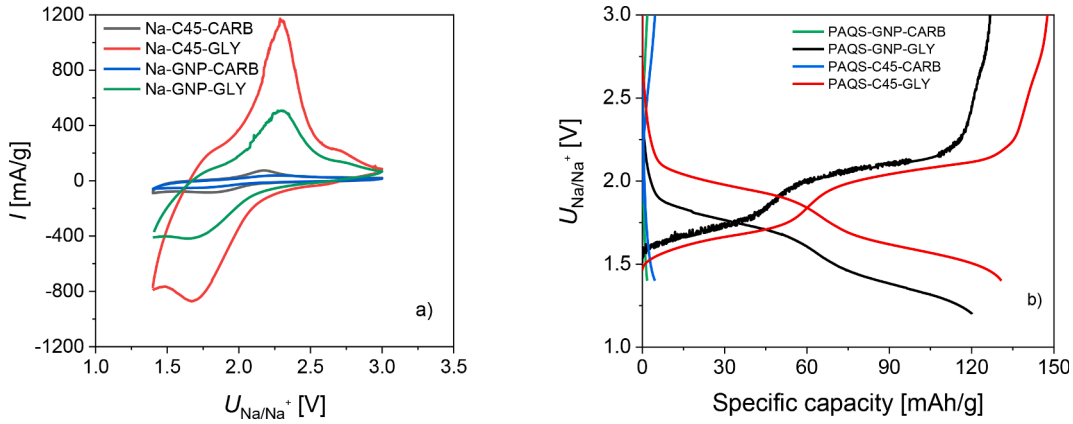

Fig. 5. (A) Cyclic voltammetry of PAQS-C45 and PAQS-GNP electrodes at a scan rate of 5 mV/s for glymes and carbonates type Na electrolytes, (B) Galvanostatic cycling of coin cells at 0.2 C rate for glymes- and carbonates-based type Na electrolyte for PAQS with conductive filler based on Super C45 and GNP.

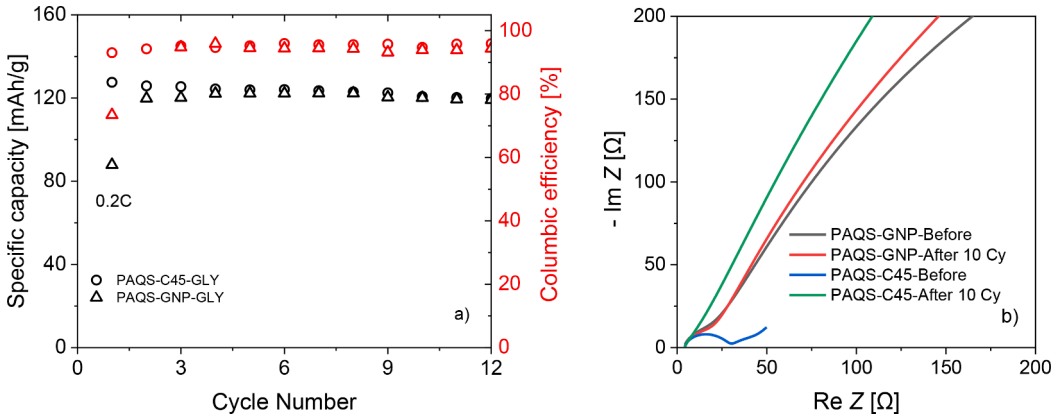

Fig. 6. (A) Galvanostatic cycling of coin cell at 0.2 C rate for glymes-based type Na electrolyte for PAQS with conductive filler based on Super C45 and GNP, (B) EIS analysis for cells with glymes-based type Na electrolyte and with electrodes with conductive fillers based on Super C45 and GNP – before cycling and after 10 cycles,.

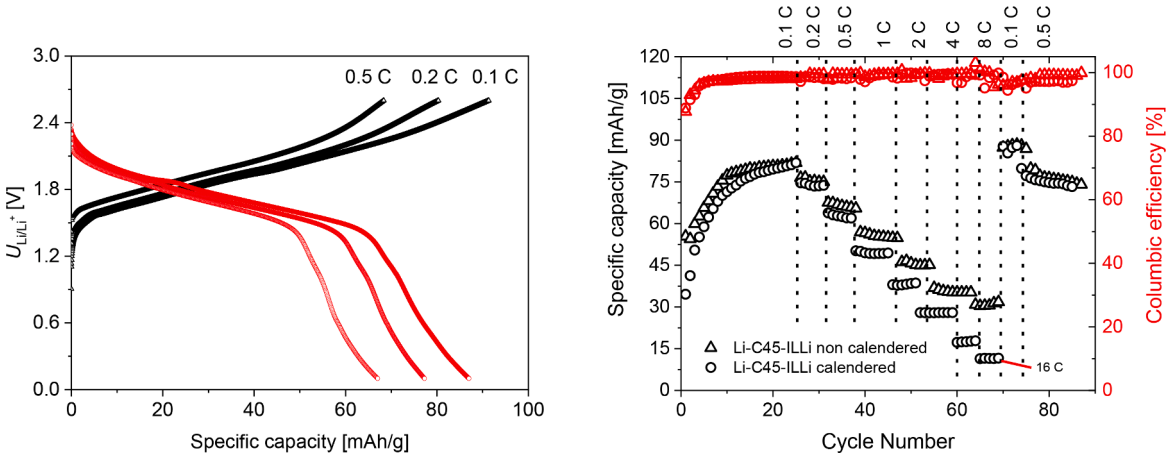

Fig. 7. (A) Long term galvanostatic cycling of coin cells (PAQS/LMO) at various cycling rates for IL-based type Li electrolyte for PAQS electrode with conductive filler based on Super C45, (B) Galvanostatic cycling of coin cells at various rates for IL-based type Li electrolyte for PAQS electrode with conductive filler based on Super C45 and influence of calendering (300 kg/cm^2) of the electrode on cell performance.

the 50th cycle (64 mAh/g). It means the average capacity decrease was 0.05% per cycle. This is a very good result for a cell with OCV higher than 2.0 V assembled at ambient conditions.

3.4. R2R printed secondary batteries assembled at ambient conditions with an electrolyte based on ionic liquid

Fully printed cells based on the electrode pair PAQS/LMO were manufactured in the R2R regime of printing except for cell assembly and electrolyte printing (Fig. 9). The gel-type electrolyte was used. It is based

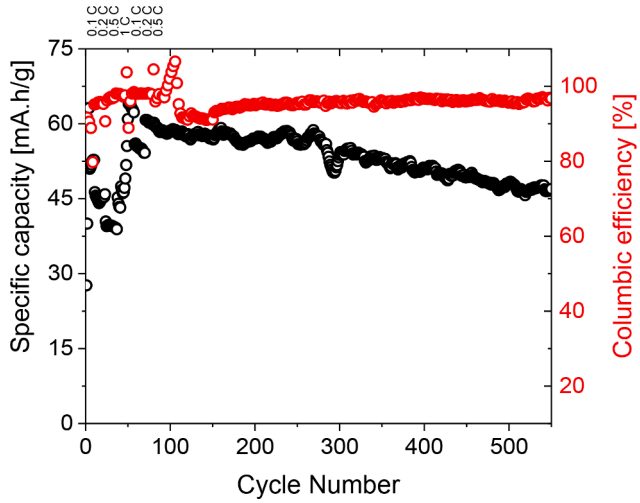

Fig. 8. Long-term galvanostatic cycling of coin cell (PAQS/LMO) at various rates (mostly 0.5C) for IL-based type Li electrolyte for non calendered PAQS electrode with conductive filler based on Super C45.

on 1 M LiTFSI in IL gelled by the addition of an inorganic gelling agent in the amount of 8 wt.%. Glass fibre substrate was used as a separator. The whole cell was encapsulated using electrolyte-resistant double-side self-adhesive tape, together with two-component epoxy glue deposited by microdispensing technique.

The active electrode area of the assembled cells is approximately 12 cm^2. The assembled cell was cycled at a cycling rate of 0.2C. We can see from Fig. 10a that the charge/discharge characteristic has softer progress compared to the coin cell equivalent. This is caused by the overall lower conductivity of current collectors, bigger gaps between electrodes in the fully printed cell, and by lower downforce in comparison to the coin cell. Growth of capacity during the initialization process is evident from Fig. 10b. The obtained specific capacity is also lower in comparison to the coin cell equivalent and its maximum value was 57 mAh/g, which means an area capacity of about 0.35 mAh/cm^2. The specific capacity was about 22% lower than in the case of the coin cell (73.1 mAh/g). It is clear from galvanostatic cycling at a rate of 0.2C that at the 100th cycle the specific capacity is about 39% lower than the maximum.

4. Conclusions

The operation of printed electronic devices generally requires small energy sources and so in many cases, secondary batteries are beneficial because they ensure reusability. The printed batteries have also other benefits, where one of the most significant is the possibility to manufacture the whole devices (Smart Labels, IoT devices, sensors, etc.) at the same time in a printing line together with the battery. This approach has production and economic advantages because connecting and assembling conventional high effective pouch cells is rather problematic and final flexibility and thickness is mediocre as well. In our study, we demonstrated the organic-based electrodes based on poly (1,5-anthraquinonyl sulfide) which is possible to fabricate by screen- or stencil-printing. It was proven, that it is possible to use these electrodes as cathodes in LIB or SIB secondary batteries when assembled in an inert atmosphere. In the case of usage of PAQS electrodes in LIB, it was found, that glymes- and IL-based electrolytes allowed the very good performance of prepared cells. A specific capacity of up to 124.4 mAh/g was obtained for cells assembled in Ar atmosphere with glymes-based

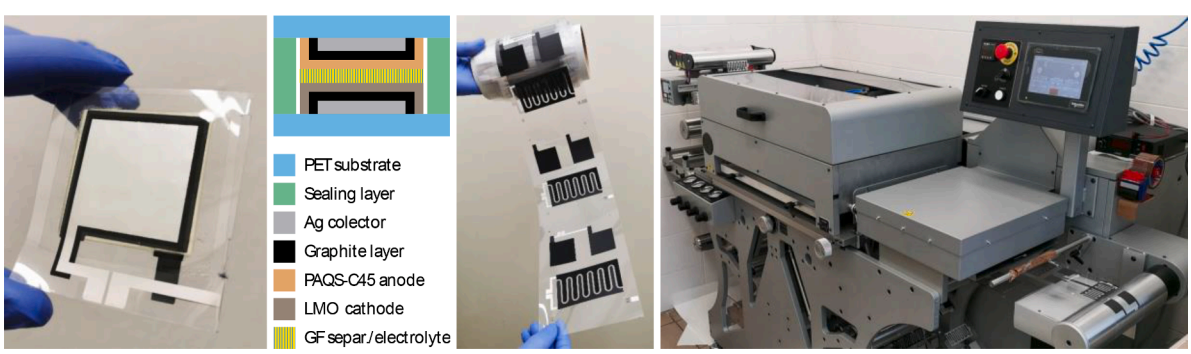

Fig. 9. (A) R2R fully printed PAQS/LMO battery, Structure of the layers of R2R printed battery (B) Roll of printed battery with electrodes PAQS/LMO, (C) R2R screen printing of PAQS/LMO secondary batteries.

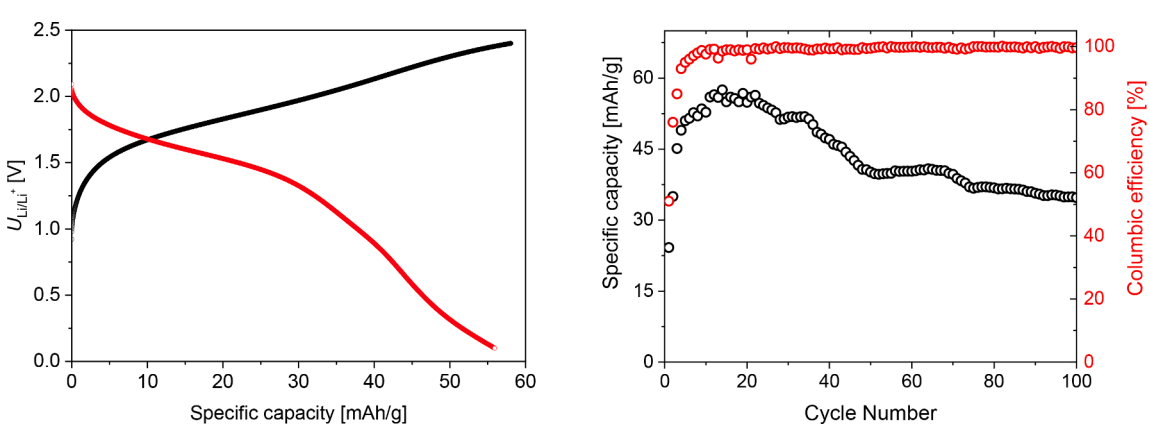

Fig. 10. (A) Charging/discharging curve for R2R printed battery for IL-based type Li electrolyte for PAQS with conductive filler based on Super C45, (B) Long term galvanostatic cycling of R2R printed battery at a cycling rate of 0.2 C for IL-based type Li electrolyte for PAQS with conductive filler based on Super C45.

electrolyte at a cycling rate of 0.2C. A specific capacity of up to 88 mAh/g was reached for coin cells based on the electrode pair PAQS/LMO with the electrolyte based on LiTFSI and IL as a solvent. It must be noted, that the cell assembled at ambient conditions was able to operate for more than 550 cycles with a decline of only 22% compared to the maximum value of specific capacity. The developed PAQS ink formulation was used for R2R fabrication of fully printed secondary batteries, which enabled OCV of more than 2 V and allowed a specific area capacity of 0.35 mAh/cm^2 at 0.2C or a specific capacity of 57 mAh/g. The capacity of the printed battery was 4 mAh which is suitable for powering of small devices as NFC smart labels, electrochromic display, etc. Finally, it was demonstrated that it is possible to use the printed PAQS electrodes in SIB where the results obtained for cells assembled at inert conditions were slightly better than for LIB equivalents in terms of specific capacity (127.6 mAh/g) for PAQS-C45 at a cycling rate of 0.2C.

Authors statement

Tomas Syrovy was responsible for strategic orientation of battery cells development; he was responsible for ink formulations development and designing of technology processes of fabrication of the cells including R2R process of its manufacturing. He ensured measurement and analysis of obtained data for cells based on IL electrolyte including R2R printed cells. He was writing main parts of article and he analysed obtained data.

Tomas Kazda was responsible for characterisation of cells and electrodes for LIB and SIB assembled in glove box. He was writing discussion of article and he analysed obtained data.

Lucie Syrova covered preparation of selected LIB cells and assembling of R2R printed cells. She ensured measurement of selected parameters of functional layers.

Jiri Akrman synthetized PAQS derivates including optimization of synthesis in larger amounts.

Declaration of Competing Interest

None.

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References

[1] P. Andersson Ersman, J. Kawahara, M. Berggren, Printed passive matrix addressed electrochromic displays, Organic Electron 14 (2013) 3371–3378.
[2] A. Martínez-Olmos, J. Fernández-Salmerón, N. Lopez-Ruiz, A. Rivadeneyra Torres, L. Capitan-Vallvey, A. Palma, Screen printed flexible radiofrequency identification tag for oxygen monitoring, Anal. Chem. 85 (2013) 11098–11105.
[3] S. Ali, S. Khan, A. Bermak, Inkjet-printed human body temperature sensor for wearable electronics, IEEE Access 7 (2019) 163981–163987.
[4] C. Costa, R. Gonçalves, S. Lanceros-Méndez, Recent advances and future challenges in printed batteries, Energy Storage Mater. 28 (2020) 216–234.
[5] J. Kim, R. Kumar, A. Bandodkar, J. Wang, Advanced materials for printed wearable electrochemical devices: a review, Adv. Electron. Mater. 3 (2017) 15.
[6] C. Du, Q. Liang, Y. Luo, Y. Zheng, Q. Yan, Recent advances in printable secondary batteries, J. Mater. Chem. A 5 (2017) 22442–22458.
[7] K. Choi, D. Ahn, S. Lee, Current status and challenges in printed batteries: toward form factor-free, monolithic integrated power sources, ACS Energy Lett. 3 (2017) 220–236.
[8] J. Geng, J. Bonnet, S. Renault, F. Dolhem, P. Poizot, Evaluation of polyketones with N-cyclic structure as electrode material for electrochemical energy storage: case of tetraketopiperazine unit, Energy Environ. Sci. 3 (2010) 1929–1933.
[9] Z. Song, H. Zhan, Y. Zhou, Anthraquinone based polymer as high performance cathode material for rechargeable lithium batteries, Chem. Commun. 1 (2009) 448–450.
[10] M. Yao, H. Senoh, S. Yamazaki, Z. Siroma, T. Sakai, K. Yasuda, High-capacity organic positive-electrode material based on a benzoquinone derivative for use in rechargeable lithium batteries, J. Power Sources 195 (2010) 8336–8340.
[11] M. Yao, H. Senoh, T. Sakai, T. Kiyobayashi, Pentacenetetrone as a high-capacity organic positive-electrode material for use in rechargeable lithium batterie, Int. J. Electrochem. Sci. 5 (2011) 2905–2911.
[12] Z. Lei, W. Wei-kun, W. An-bang, Y. Zhong-bao, C. Shi, Y. Yu-sheng, A. MC, AQ parasitic composite as cathode material for lithium battery, J. Electrochem. Soc. 158 (2011).
[13] Y. Liang, Z. Tao, J. Chen, Organic electrode materials for rechargeable lithium batteries, Adv. Energy Mater. 2 (2012) 742–769.
[14] Y. Hanyu, Y. Ganbe, I. Honma, Application of quinonic cathode compounds for quasi-solid lithium batteries, J. Power Sources 221 (2013) 186–190.
[15] E. Quartarone, P. Mustarelli, Electrolytes for solid-state lithium rechargeable batteries: recent advances and perspectives, Chem. Soc. Rev. 40 (2011) 2525–2540.
[16] S. Tobishima, J. Yamaki, A. Yamaji, Cathode Characteristics of Organic Electron Acceptors for Lithium Batteries, J. Electrochem. Soc. 131 (1984) 57–63.
[17] A. Kukla, Y. Shirshov, S. Piletsky, Ammonia sensors based on sensitive polyaniline films, Sensor. Actuator. B 37 (1996) 135–140.
[18] P. Bu, S. Liu, Y. Lu, S. Zhuang, H. Wang, F. Tu, Effects of carbon black on the electrochemical performance of lithium-organic coordination compound batteries, Int. J. Electrochem. Sci. 7 (2012) 4617–4624.
[19] A. Reddy, S. Nagarajan, P. Chumyim, S. Gowda, P. Pradhan, S. Jadhav, M. Dubey, G. John, P. Ajayan, Lithium storage mechanisms in purpurin based organic lithium ion battery electrodes, Sci. Rep. 2 (2012) 5.
[20] W. Wan, H. Lee, X. Yu, C. Wang, K. Nam, X. Yang, H. Zhou, Tuning the electrochemical performances of anthraquinone organic cathode materials for Li-ion batteries through the sulfonic sodium functional group, RSC Adv. 4 (2014) 19878–19882.
[21] W. Xu, A. Read, P. Koech, D. Hu, C. Wang, J. Xiao, A. Padmaperuma, G. Graff, J. Liu, J. Zhang, Factors affecting the battery performance of anthraquinone-based organic cathode materials, J. Mater. Chem. 22 (2012) 4032–4039.
[22] Y. Li, L. Liu, C. Liu, Y. Lu, R. Shi, F. Li, J. Chen, Rechargeable aqueous polymer-air batteries based on Polyanthraquinone anode, Chem 5 (2019) 2159–2170.
[23] J. Bitenc, K. Pirnat, T. Bančič, M. Gaberšček, B. Genorio, A. Randon-Vitanova, R. Dominko, Anthraquinone-based polymer as cathode in rechargeable magnesium batteries, ChemSusChem 8 (2015) 4128–4132.
[24] B. Tian, G. Ning, W. Tang, C. Peng, D. Yu, Z. Chen, Y. Xiao, C. Su, K. Loh, Polyquinoneimines for lithium storage: more than the sum of its parts, Mater. Horiz. 3 (2016) 429–433.
[25] B. Genorio, K. Pirnat, R. Cerc-Korosec, R. Dominko, M. Gaberscek, Electroactive organic molecules immobilized onto solid nanoparticles as a cathode material for lithium-ion batteries, Angewandte Chemie Int. Ed. 49 (2010) 7222–7224.
[26] Y. Yu, Binding energy and work function of organic electrode materials phenanthraquinone, pyromellitic dianhydride and their derivatives adsorbed on graphene, ACS Appl. Mater. Interfaces. 6 (2014) 16267–16275.
[27] F. Xu, S. Jin, H. Zhong, D. Wu, X. Yang, X. Chen, H. Wei, R. Fu, D. Jiang, Electrochemically active, crystalline, mesoporous covalent organic frameworks on carbon nanotubes for synergistic lithium-ion battery energy storage, Sci. Rep. 5 (2015) 6.
[28] I. Sultana, M. Rahman, J. Wang, C. Wang, G. Wallace, H. Liu, Indigo carmine (IC) doped polypyrrole (PPy) as a free-standing polymer electrode for lithium secondary battery application, Solid State Ionics 215 (2012) 29–35.
[29] T. Ma, Q. Zhao, J. Wang, Z. Pan, J. Chen, A sulfur heterocyclic quinone cathode and a multifunctional binder for a high-performance rechargeable lithium-ion battery, Angewandte Chemie Int. Ed. 55 (2016) 6428–6432.
[30] M. Sterby, R. Emanuelsson, X. Huang, A. Gogoll, M. Strømme, M. Sjödin, Characterization of PEDOT-quinone conducting redox polymers for water based secondary batteries, Electrochim. Acta 235 (2017) 356–364.
[31] T. Syrový, T. Kazda, L. Syrová, J. Vondrák, L. Kubáč, M. Sedlaříková, Cathode material for lithium ion accumulators prepared by screen printing for Smart Textile applications, J. Power Sources 309 (2016) 192–201.
[32] S. Lawes, Q. Sun, A. Lushington, B. Xiao, Y. Liu, X. Sun, Inkjet-printed silicon as high performance anodes for Li-ion batteries, Nano Energy 36 (2017) 313–321.
[33] Y. Zhao, Q. Zhou, L. Liu, J. Xu, M. Yan, Z. Jiang, A novel and facile route of ink-jet printing to thin film SnO2 anode for rechargeable lithium ion batteries, Electrochim. Acta 51 (2006) 2639–2645.
[34] Y. Zhao, G. Liu, L. Liu, Z. Jiang, High-performance thin-film Li4Ti5O12 electrodes fabricated by using ink-jet printing technique and their electrochemical properties, J. Solid State Electrochem. 13 (2009) 705–711.
[35] Y. Zhao, J. Xu, L. Liu, J. Yang, Z. Jiang, Thin film LiCoO2 Cathode prepared by using ink-jet printing technique and its electrochemical properties, Chem. J. Chine. Univer.-Chine. 28 (2007) 1122–1125.
[36] M. Park, S. Hyun, S. Nam, Characterization of a LiCoO2 thick film by screen-printing for a lithium ion micro-battery, J. Power Sources 159 (2006) 1416–1421.
[37] C. Jung, T. Zhao, L. An, L. Zeng, Z. Wei, Screen printed cathode for non-aqueous lithium–oxygen batteries, J. Power Sources. vol. 297 (2015) 174–180.